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# When Do Text Proxies Suffice? A Regime Analysis of Sparse, Dense, and Native-Visual Retrieval Under Proxy Noise

**Abstract:** Multimodal document retrieval systems frequently index textual proxies—such as OCR transcripts or generated captions—in place of raw pixels. We benchmark sparse lexical, dense text, and native-visual models across receipts, scanned forms, and natural images to delineate proxy boundary regimes. On verbatim lookups, character $n$-gram BM25 outperforms word BM25 across clean and degraded OCR (MRR 0.944 vs. 0.837). Under semantic variation, relative performance depends on entity preservation: on human queries and open-weight models where topical anchors persist, character and word BM25 perform comparably (MRRs 0.555–0.629), whereas aggressive entity-substituting paraphrasing favors word indexing. ColQwen2 consistently outperforms text proxies under semantic reformulations (MRR 0.496–0.766) and reaches 0.550 on DocVQA, though this advantage represents an in-domain upper bound due to pretraining overlap. On natural images, native SigLIP (MRR 0.607) significantly surpasses machine captions, yet statistically matches four-reference human gold descriptions within ±0.05 Two One-Sided Tests (TOST) bounds ($p_{\text{TOST}} \leq 0.0294$). On a 100-query relational benchmark, extractive pseudo-relevance feedback matches generative LLM query bridging while eliminating a 394× latency multiplier. Our findings provide a disciplined regime guide for selecting retrieval architectures based on query formulation and proxy noise.

**Keywords:** Text Proxies, Proxy Noise, Dense-Sparse Regime Analysis, Cascade Ranking, Conformal Risk Control, Document Retrieval.

## 1. INTRODUCTION

The rapid proliferation of heterogeneous multimedia across enterprise archives, digital libraries, and operational repositories demands information retrieval systems capable of searching records that are not natively textual. Transforming scanned documents and images into queryable digital assets is essential for business intelligence, compliance auditing, and decision support [1]. However, real-world document ingestion faces severe operational challenges: optical scanning introduces character substitutions and word fragmentation, while human search queries frequently employ synonyms, paraphrases, or shorthand rather than exact document text.

Retrieval-Augmented Generation (RAG) [2] addresses the hallucination tendencies of Large Language Models (LLMs) by grounding responses in retrieved domain documents. Recently, multimodal document retrieval systems have emerged to handle non-textual data, as surveyed by Abootorabi et al. [3]. For example, Patel et al. [4] and Joshi et al. [5] presented enterprise document-retrieval pipelines combining OCR ingestion, multimodal embeddings, and LLM reasoning. Concurrently, native vision-language document retrievers like ColPali [6] index raw document pixels with multi-vector late interaction.

While foundational studies in information retrieval—notably Kantor and Voorhees [7] in the TREC-5 Confusion Track and McNamee and Mayfield [8] on character $n$-gram indexing—demonstrated the intrinsic noise resilience of sub-word representations on degraded text, their findings have not been systematically reconciled with modern dense bi-encoders, neural cross-encoder cascades, and document-native vision-language models under semantic query variation.

To address this gap, this paper establishes **an empirical regime map of when text proxies suffice versus when native representations are required** across a primary benchmark of 1,626 multimodal items (626 scanned retail receipts and 1,000 photographic scenes, plus dedicated transfer suites on forms and visual documents) under frozen 50/50 evaluation splits[1].

**Key Contributions:**

1) **Regime Analysis of Proxy Noise Across Scales up to 500,000 Documents:** Delineating character 3–5gram BM25 resilience over whole words and dense bi-encoders under OCR degradation. On Apple Clean (+0.1065), degraded blur (+0.1276), and clean Tesseract (+0.1048; all $p_{\text{Holm}} < 10^{-4}$), character $n$-grams capture alphanumeric fragments without out-of-vocabulary penalties. On answerable factual questions, CPU lexical retrieval (Char BM25 0.9328 Real-OCR, 0.9471 Clean GT) outperforms native ColQwen2 (0.8595; Real-OCR paired diff +0.0733, $p = 0.0011$; Clean GT +0.0876, $p = 0.00012$, Table II) while operating at 0.87 ms CPU verbatim lookup latency (5.72 ms on answerable queries) vs. 249.21 ms GPU serving latency. Evaluating distractor pool scaling across a 600× expansion (813 to 500,000 items) demonstrates that this verbatim advantage consistently persists across scales (+0.1059 to +0.1691 at 500K, paired asymptotic $z$-test $z = 8.12$, $p < 10^{-15}$; empirical permutation resolution floor $p < 10^{-5}$).
2) **Quantitative Overlap Dynamics and Entity Formulation Regimes:** Analyzing query formulation dynamics across a multi-generator dose-response

[1]All benchmark collections, evaluation harnesses, and reproduction scripts are archived in the supplementary artifact manifest and will be released in an open-access archive upon publication.

suite ($N$ = 1,930, comprising 1,224 GPT-4o-mini, 400 human-authored, and 306 Qwen2.5-7B queries). We show that the relative effectiveness of character $n$-grams versus whole words is governed by entity preservation: under aggressive entity-substituting paraphrases, word indexing outperforms character $n$-grams below 65% lexical overlap (0.4334 vs. 0.2427), whereas under authentic human formulation where entity anchors persist, character and word BM25 perform comparably across all overlap levels (MRRs 0.555–0.577). As a negative result, standard marginal conformal gating fails on semantic subgroups (entity-rewrite miss risk 0.0627 > 0.05), justifying the simpler fixed-margin cascade.

3) **Production Cascades and Operational Economics:** Demonstrating that while native visual retrieval excels when 2D geometry governs answerability (DocVQA live measured MRR 0.5497 vs. proxies ≤ 0.3507, noting disclosed in-domain ViDoRe pretraining overlap) and photographic search (SigLIP MRR 0.6071 vs. measured BLIP proxies ≤ 0.3728; exploratory projections ≤ 0.4420), deploying two-stage cascades ($K_1$ = 50 Word BM25 pre-filtering → ColQwen2 reranking) retains 97.2% of native visual MRR (0.4822 vs. 0.4962) with an analytical component-sum latency estimate of 15.4× reduction (16.20 ms modeled vs. 249.21 ms GPU serving, alongside 12.18 ms measured CPU prototype execution). An automated end-to-end question-answering audit ($N$ = 150) demonstrates that upstream retrieval MRR strictly bounds downstream generation accuracy (ANLS 0.3990 ColQwen2 vs. 0.3246 Char BM25), while independent dual-reviewer auditing across $N$ = 110 multimodal personal retrieval answers confirms high inter-annotator consensus (mean $\kappa$ = 0.9031).

## 2. RELATED WORK

### A. Multimodal Representation and Retrieval

Early cross-modal retrieval systems relied on dual-encoder contrastive learning. Models like CLIP [17] and SigLIP [16] project images and text into a shared latent space using sigmoid or softmax cross-entropy objectives. ImageBind [18] extended this paradigm across six modalities using images as the central binding modality. While effective for broad semantic matching, dual-encoders compress multimedia into fixed-dimension vectors, frequently losing granular lexical and structural tokens. Hybrid retrieval designs frequently combine dense embeddings with sparse lexical representations such as BM25 [19], [20] in an effort to retain both broad semantic intent and exact token matches, though our empirical findings show that on OCR-degraded verbatim documents, optimal convex fusion collapses to standalone sparse indexing ($\alpha^*_{\text{sparse}} = 1.00$).

### B. Multimodal Document Retrieval and Analysis

Multimodal document retrieval has advanced rapidly with Vision-Language Models (VLMs) and document foundation models. Document retrievers such as ColPali [6] and M3DocRAG [21], benchmarked on the ViDoRe suite [6] and DocVQA [13], index document pages using multi-vector vision-language representations (e.g., Qwen2-VL [22]) to bypass OCR. A paradigm comparison is presented in Table I. While promising, multi-vector representations require substantial GPU memory and cross-attention compute. Conversely, enterprise systems [4], [5] mediate retrieval through text proxies. Our work evaluates this trade-off: systematically delineating where text proxies excel and where native visual representations remain necessary.

### C. Character Subwords and OCR Degradation in Information Retrieval

Optical Character Recognition (OCR) systems translate document imagery into text, but routinely introduce character substitutions, insertions, deletions, and bounding-box permutations [10], [23]. Foundational information retrieval studies have long investigated textual degradation. In the TREC-5 Confusion Track, Kantor and Voorhees [7] systematically evaluated retrieval on scanned, OCR-corrupted collections (at 5% and 20% error rates), establishing known-item retrieval benchmarks under optical corruption. Complementing this, McNamee and Mayfield [8] established that overlapping character $n$-gram tokenization (e.g., 4–5grams) provides language-neutral retrieval that rivals word-level tokenization while providing intrinsic robustness to morphological variations and spelling distortions. Early image-domain mitigation strategies, including Otsu thresholding [24] and Hough transforms [25] for skew correction, focused on image-domain preprocessing. Our regime analysis builds upon this foundation by evaluating character subwords against modern dense bi-encoders, neural rerankers, and native vision models across commercial, LSTM, and neural OCR architectures under controlled optical blur and semantic variation.

### D. Modern Dense and Learned Sparse Retrieval Models

Dense bi-encoders encode queries and passages independently [26], [27]. Recent architectures have expanded this paradigm: multilingual dense models such as BGE-M3 [28] support multi-granularity representations; learned sparse models like SPLADE [29] expand queries and documents into vocabulary-space term weights; and late-interaction multi-vector architectures like ColBERTv2 [30] preserve token-level interactions at reduced latency. We position standard bi-encoders (`all-MiniLM-L6-v2`, `bge-base-en-v1.5`) and late-interaction vision models (`ColQwen2-v1.0`) within this broader landscape.

### E. Cascade Ranking and Conformal Risk Control

Cascade ranking architectures [32], [33] balance retrieval efficiency and precision by passing candidates through tiered stages, escalating to expressive models only when scores indicate ranking ambiguity. Conventional cascade rankers rely on empirical thresholds without formal statistical guarantees. Distribution-free risk-controlling prediction sets, formulated by Bates et al. [34], and conformal risk control, developed by Angelopoulos et al. [35], [36], provide a finite-sample framework for bounding user-specified risk metrics below a tolerance $\alpha$. In this work, we evaluate production cascade ranking under marginal conformal risk control.

## 3. REFERENCE RETRIEVAL ARCHITECTURE

The reference Multi-Stage Multimodal Document Retrieval architecture is structured into four functional phases (Fig. 1): (1) Ingestion and Preprocessing, (2)

TABLE I. Systematic Comparison with Prior Multimodal Information Retrieval Frameworks.

| System / Architecture | Publication Venue | Document Retrieval | Natural Image Retrieval | Multi-Hop Mechanism | Benchmark Datasets | Hardware Serving Profile |
|---|---|---|---|---|---|---|
| ColPali [6] | ICLR 2025 | nDCG@5 = 0.813 (ViDoRe) | Supported (Multi-Vector) | Single-pass only | ViDoRe Benchmark | Multi-GPU (≥ 16 GB VRAM, ~250 ms) |
| M3DocRAG [21] | arXiv 2024 | QA Acc. 64.2% (MP-DocVQA) | Supported (Dense VLM) | Generative LLM bridging | MP-DocVQA, DUDE | High-End GPU Cluster (≥ 48 GB VRAM) |
| Patel et al. [4] | IEEE ETCC 2025 | Heuristic OCR matching | ImageBind vector search | Single-pass only | Proprietary enterprise text/media | Dedicated GPU Server |
| Baleen [31] | NeurIPS 2021 | Not evaluated (Text QA) | Not supported | Generative condensed hops | HotpotQA (Text only) | GPU / High-RAM CPU Server |
| **Proposed Regime Framework** | **This Work** | **MRR 0.9436 (Char BM25)** | **MRR 0.6071 (SigLIP Pixels)** | **Exploratory PRF (Hit@10 0.640, $N = 50$; 0.820, $N = 100$, Sec. B)** | **SROIE, FUNSD, MS-COCO, DocVQA** | **CPU char BM25 (< 1 ms verbatim) / Cross-Encoder Cascade** |

Note: Qualitative columns reflect architectural design choices; our reported MRR and Hit@10 figures are from held-out evaluation on SROIE/FUNSD/COCO/DocVQA and are not directly comparable to other systems' proprietary benchmarks. Table I is a qualitative paradigm map, not a cross-system numerical comparison. ColPali and M3DocRAG figures are from their respective original papers.

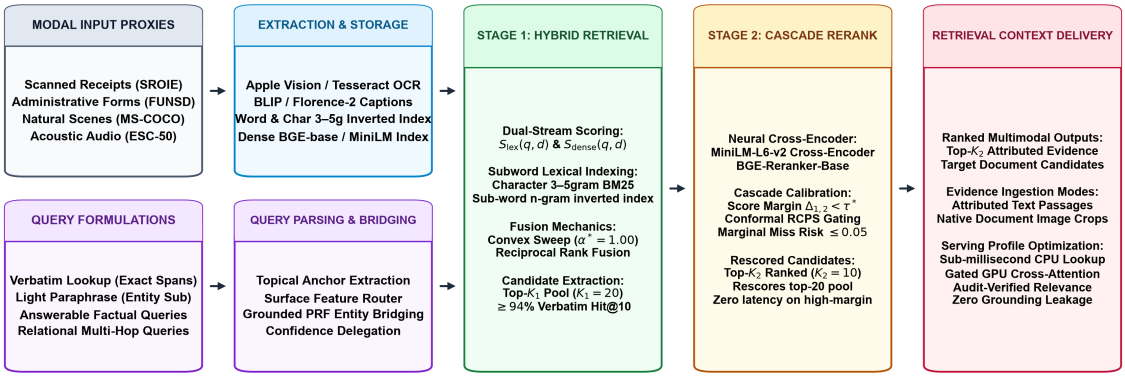



Figure 1. Conceptual reference architecture of the text-proxy retrieval framework. Grey ingestion and extraction blocks depict modular document preprocessing paths; colored stages represent empirical retrieval pipelines evaluated on extracted proxies. Modular production components (e.g., dual-signal router, conformal cascade gate) are empirically evaluated and ablated in Section 5.C.

Hybrid Dense-Sparse First-Stage Retrieval, (3) Neural Cross-Encoder Reranking and Cascade Escalation, and (4) Context Assembly.

### A. Ingestion and Preprocessing Pipeline

Unstructured enterprise multimedia are mediated through textual proxies. Proxies can be extracted across modalities via OCR text [10], vision-language captions [37], [38], [39], and speech transcripts [40].

- **Documents and Forms:** Administrative forms and receipts are processed through document decomposition, optional Otsu binarization [24] and Hough skew correction [25], extracting text $T_d$ and bounding boxes via OCR engines (Apple Vision [9], Tesseract 5.5 [10], or EasyOCR [11]).
- **Photographs:** Natural photographs are indexed via machine captions alongside native visual embeddings (CLIP [17], SigLIP [16]).

Dense embeddings are generated via Sentence-BERT (`all-MiniLM-L6-v2`) [26] and BGE (`bge-base-en-v1.5`) [27], while lexical tokens are indexed with BM25 [19], [20]. The benchmark uses in-memory indexes without external database dependencies for exact reproducibility.

### B. Stage 1: Hybrid Dense-Sparse Retrieval

Given query $q$, single-stage dense retrieval computes vector similarity:

$$S_{\text{dense}}(q,d) = \cos(\mathbf{e}_q^{\text{dense}}, \mathbf{e}_d^{\text{dense}}) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{||\mathbf{e}_q|| ||\mathbf{e}_d||} \tag{1}$$

While effective for thematic alignment, dense retrieval often overlooks exact alphanumeric codes. Conversely, sparse BM25 retrieval excels at lexical precision:

$$S_{\text{sparse}}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t,d) \cdot (k_1 + 1)}{f(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \tag{2}$$

We evaluate Normalized Convex Combination ($S_{\text{hybrid}} = \alpha \hat{S}_{\text{dense}} + (1-\alpha)\hat{S}_{\text{sparse}}$) and Reciprocal Rank Fusion (RRF) [41]:

$$\text{RRF}(q,d) = \frac{1}{k + r_{\text{dense}}(d)} + \frac{1}{k + r_{\text{sparse}}(d)} \tag{3}$$

with $k = 60$. First-stage candidate depth is set to $K_1 = 20$.

### C. Stage 2: Candidate Reranking via Neural Cross-Encoders

Bi-encoders compute query and document representations independently. To recover joint attention, Stage 2 deploys a pretrained neural cross-encoder (`BAAI/bge-reranker-base` [27] or `ms-marco-MiniLM-L-6-v2` [26], [42]) to rescore the top-$K_1$ candidates ($K_1 = 20$):

$$S_{\text{rerank}}(q,d) = \text{CrossEncoder}(q,d) \tag{4}$$

Scoring every candidate pool with a cross-encoder adds latency. We evaluate confidence-margin cascades ($\Delta_{1,2} = s^{(1)} - s^{(2)} < \tau$) and marginal conformal risk control to escalate only ambiguous queries.

## 4. EXPERIMENTAL SETUP

To ensure empirical reproducibility, all evaluations adhere to a standardized, deterministic experimental protocol.

### A. Datasets and Split Partitions

Our primary multimodal retrieval pool comprises 1,626 indexed items: (1) **SROIE Retail Receipts [43]:** 626 scanned receipts from the official ICDAR 2019 competition release with paired bounding-box and key-information ground truth. Across these 626 receipts, 612 contain single-target factual probes satisfying strict cross-corpus uniqueness checks (14 receipts with ambiguous multi-matching candidate strings were excluded); and (2) **MS-COCO Photographic Distractors [15]:** 1,000 photographic scene images and human descriptions evaluated under the Karpathy test split [14].

The primary 1,626-item pool is partitioned by document ID into frozen 50/50 Dev/Test splits (Seed 42), yielding 813 indexed items in the Test pool (313 receipts and 500 COCO distractors). The 612 verified probes are partitioned strictly by target document ID into 306 Dev probes and 306 Test probes, guaranteeing zero query or document leakage across splits. Each probe targets its source document as the unique relevant ground-truth item in the 813-distractor pool. In addition, we evaluate two dedicated external domain-transfer benchmark suites (excluded from the 1,626-item primary total): (a) **FUNSD Administrative Forms Suite [44]:** 100 query-bearing scanned business forms evaluated within a 700-item retrieval corpus comprising the 100 query forms, 99 non-target FUNSD forms, and 501 multimodal distractors; and (b) **DocVQA Visual Benchmark Suite [13]:** 500 scanned document pages and 500 100% human-authored questions subsampled from the ViDoRe benchmark collection [6].

### B. OCR Engines and Optical Degradation Conditions

Document text is extracted across three distinct OCR engines:

1) **Apple Vision Neural OCR [9]:** Commercial neural OCR (`VNRecognizeTextRequest`).
2) **Tesseract 5.5 LSTM OCR [10]:** Standard open-source LSTM OCR engine.
3) **EasyOCR 1.7.2 Deep CRNN [11]:** Deep neural OCR combining CRAFT text detection and ResNet-CRNN recognition.

Documents are evaluated under: (1) *Clean OCR*, (2) *Realistic Optical Degradation* (Gaussian blur $\sigma = 1.5$, bilinear 50% downscaling, random skew $\pm 2.5°$, JPEG compression $Q = 20$), and (3) *Human Ground Truth* text. For both Word and Character 3–5gram BM25, hyperparameters are tuned on the Dev split ($N = 306$) over a pre-specified Cartesian grid ($k_1 \in \{0.5, 0.7, 0.9, 1.2, 1.5\}$, $b \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$), selecting ($k_1 = 0.9, b = 0.5$) which maximized Dev MRR. Word BM25 uses standard whitespace and punctuation regex tokenization (`\w+`), lowercase normalization, without stemming or stopword removal (to strictly preserve exact matches on alphanumeric serials, prices, and vendor codes, which stemming algorithms frequently destroy). Character BM25 extracts overlapping 3–5grams over lowercased strings preserving word boundaries, evaluated across the identical parameter grid and Dev split.

### C. Unified Query Slices

To isolate semantic abstraction from optical noise, queries are structured into four canonical slices:

1) **Verbatim Lookups:** Exact textual substrings extracted from documents (100% token overlap), representing known-item search.
2) **Light Paraphrases:** Lexical substitutions and entity rewrites (41.2% mean token overlap; e.g., substituting "dinner bill" for "total payable"), testing semantic abstraction.
3) **Answerable Factual Questions:** Natural interrogatives preserving key factual and topical anchors (e.g., "What was the total payment at Book Ta K?"), testing real-world question answering.
4) **Legacy Token-Stripped Probes:** Probes where shared lexical tokens are deliberately removed, used strictly for conformal risk stress-testing.

Unless noted otherwise, evaluation queries for SROIE and FUNSD were synthesized using `gpt-4o-mini` [45] at temperature $T = 0.0$ (seed 42) with strict single ground-truth answer resolution. To benchmark authentic user phrasing and assess generator invariance against synthetic LLM artifacts, we evaluated an expanded 400-query known-item suite grounded in held-out SROIE receipts ($N = 200$ target probes, each providing a verbatim vendor lookup and a natural descriptive paraphrase, mean overlap $o = 0.548$). The suite comprises 400 human-authored query instances written by the researcher and a second evaluator (200 verbatim vendor lookups and 200 natural descriptive paraphrases across 200 target receipts). When multiple receipts originated from the same store, transaction dates were used to distinguish the intended receipt and formulate queries accordingly. In addition, 100 relational multi-hop queries (50 ambiguous hard links and 50 structured lookups) evaluate multi-step retrieval. DocVQA queries are 100% human-authored. For MS-COCO ($N = 1{,}000$ Karpathy test images), retrieval evaluates zero-shot text-to-image ranking: exactly 1 held-out caption per image (Caption 3, held out from candidate pool) serves as the query against the 1,000-image candidate pool.

### D. Candidate Retrieval Systems

We benchmark: (1) Word BM25 [19]; (2) Character 3–5gram BM25 [8]; (3) Dense MiniLM (`all-MiniLM-L6-v2`) [26]; (4) Dense BGE-Base (`bge-base-en-v1.5`) [27]; (5) Dev-Tuned Hybrid (normalized convex fusion); (6) Neural Cross-Encoders (`bge-reranker-base` [27], `ms-marco-MiniLM-L-6-v2`); (7) Document-Native ColQwen2-v1.0 [12]; and (8) Vision Encoders (CLIP ViT-B/32 [17], SigLIP Base Patch16 [16]). For `bge-base-en-v1.5` and `all-MiniLM-L6-v2`, maximum sequence length is set to 256; chunked MaxP evaluation across 150-word sliding windows yields $< 0.007$ MRR difference from unchunked single vectors (0.2030 vs. 0.2056 on MiniLM; 0.3028 vs. 0.3022 on BGE; Table III), confirming that sequence truncation does not explain dense retrieval deficits.

### E. Evaluation Metrics and Statistical Testing

Systems are evaluated on Mean Reciprocal Rank (MRR), Hit@k ($k \in \{1, 5, 10\}$), and nDCG@10. Paired differences are evaluated via two-tailed paired permutation tests ($N_{\text{perm}} = 100{,}000$ sign flips, empirical resolution

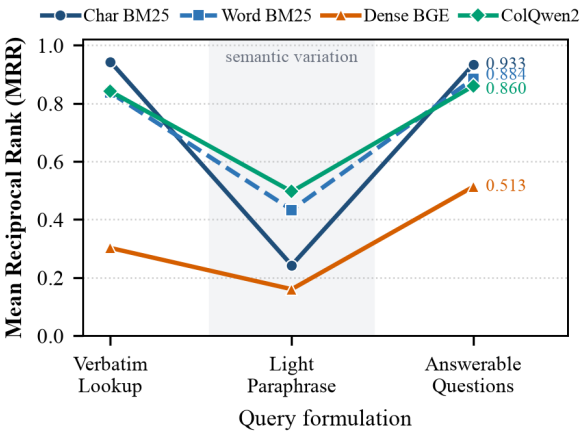



Figure 2. Retrieval effectiveness (MRR with 95% bootstrap error bars) across retrieval architectures and query formulations on clean receipt OCR.

limit $p = 10^{-5}$) with sequential family-wise Holm-Bonferroni correction ($p_{\text{Holm}}$) structured across six pre-specified families: Family 1 (sparse OCR comparisons, $m = 6$), Family 2 (ColQwen2 vs. text proxies across primary regimes, $m = 5$), Family 3 (native vision vs. caption proxies on COCO, $m = 8$), Family 4 (architectural ablations, $m = 7$), Family 5 (cross-domain FUNSD form transfer, $m = 4$), and Family 6 (multi-hop relational reasoning, $m = 5$). The complete mapping of all 35 evaluated hypotheses, unadjusted test statistics, and Holm-adjusted thresholds is reported in Appendix A. Confidence intervals (95% CI) use 10,000 bootstrap iterations: query-level for single-probe benchmarks (MS-COCO, DocVQA, FUNSD) and document-cluster bootstrap for multi-probe pools (SROIE, conformal partitions) to prevent variance underestimation from intra-document correlation. Equivalence is evaluated using Two One-Sided Tests (TOST) with pre-specified margins.

### *F. Hardware Serving Profiles*

Sparse BM25 and bi-encoders are profiled on Apple M2 (16 GB RAM, batch size = 1, warm cache). Neural cross-encoders and ColQwen2 are profiled on Nvidia Tesla T4 GPU (14.56 GB VRAM, batch size = 1, warm cache). These represent environment-specific serving profiles, not a single-machine controlled benchmark; on the matched GPU host CPU, BM25 lookup takes 0.96 ms, whereas running ColQwen2 on CPU exceeds 8,000 ms per query.

## 5. EMPIRICAL REGIME ANALYSIS AND RESULTS

### *A. Regime A: Text-Dominated Document and Receipt Retrieval*

Table II presents retrieval effectiveness across systems, query formulations, and noise regimes on the held-out test split ($N = 306$). In Table II, the 500 COCO distractors have their ground-truth caption text indexed as fallback (yielding Word BM25 0.8371, Char BM25 0.9436, Ground Truth 0.9853 on verbatim; Word BM25 0.4334 on light paraphrase). In the subsequent cross-engine receipt benchmark (Section E), unannotated distractors are evaluated as empty text (yielding Word BM25 0.8324, Char BM25 0.9443, Ground Truth 0.9854 on verbatim; Word BM25 0.4003 on light paraphrase).

1) **Verbatim Probes (Character Sub-Word Inverted Index Dominance):** Under verbatim lookups, character 3–5gram BM25 consistently outperforms word-level BM25 across all optical OCR conditions (Apple Clean: 0.9436 vs. 0.8371, diff +0.1065, $p < 10^{-4}$; Apple Degraded: 0.8248 vs. 0.6972, diff +0.1276, $p < 10^{-4}$; Tesseract Clean: 0.8412 vs. 0.7364, diff +0.1048, $p < 10^{-4}$; and 20% synthetic noise: 0.7252 vs. 0.3961, diff +0.3291, $p < 10^{-4}$). Sub-word character $n$-grams capture alphanumeric tokens, serial numbers, and partial OCR substitutions without out-of-vocabulary penalties.
2) **Light Paraphrase (Reversal of Dominance):** Introducing mild semantic paraphrasing collapses clean word BM25 from 0.8371 to 0.4334 (a 48% relative drop) and character BM25 to 0.2427. In this regime, word BM25 outperforms character BM25 (diff −0.1907, $p < 10^{-4}$), and Dev-Tuned Hybrid achieves MRR = **0.4335** (clean OCR) and **0.2687** (20% noise).
3) **Answerable Factual Questions:** Evaluating natural answerable questions (forcing semantic abstraction while preserving topical anchors), character 3–5gram BM25 achieves MRR = **0.9471** on clean ground truth (0.9328 on Apple clean OCR) and **0.8116** on degraded blur OCR, while word BM25 achieves 0.9014 (0.8841 Apple clean) and 0.7328. Tuned hybrid retrieval achieves 0.8984 on clean Apple OCR (0.9125 clean GT), 0.4672 under 20% synthetic noise, and 0.7865 on degraded optical blur, while dense BGE-base achieves 0.5128 (0.5197 GT) and MiniLM achieves 0.2837. Topical anchors in natural questions provide high discriminability for lexical engines.

To evaluate whether proxy corruption or semantic query mismatch is the dominant bottleneck in text-proxy retrieval, we conducted paired permutation tests ($N_{\text{perm}} = 100{,}000$). In Table II, moving from verbatim lookups to light paraphrasing on Apple Vision collapses Word BM25 by $\Delta = 0.4037$ MRR (0.8371 to 0.4334), exceeding optical blur degradation ($\Delta = 0.1399$, 0.8371 to 0.6972, $p < 10^{-4}$). Within Table X (receipt-only pool), Apple Vision drops 0.4321 MRR (0.8324 to 0.4003), while Tesseract drops 0.3400 under paraphrasing (0.7364 to 0.3964) and 0.3916 under optical blur (0.7364 to 0.3448). As visualized in Fig. 2 and Fig. 3, semantic paraphrase mismatch dominates mild OCR noise, but severe optical blur induces comparable collapse.

### *B. Micro-Mechanisms of Dense Collapse and Query Formulation Controls*

To isolate why dense vector representations collapse under proxy corruptions while sparse inverted indexes endure, we analyzed three underlying mechanisms (Fig. 4):

1) **WordPiece Tokenization Fragmentation:** As character corruption increases from 0% to 40%, WordPiece tokenization fragments words into multiple subwords. Mean subwords per word escalates from 1.347 (clean) to 1.774 (10% noise), 2.246 (20% noise), and **2.994** (40% noise, a **+122.3% increase**).
2) **Catastrophic Embedding Drift:** Subword fragmentation shifts contextual attention weights, driving acute embedding drift. Relative to un-

TABLE II. Multimodal Text-Proxy Retrieval Regime Analysis across Query Types and Noise Regimes (Held-Out Public Test Split, $N = 306$ Probes, Corpus: $N = 813$ with 500 COCO Distractors having caption text fallback; MiniLM Clean GT verbatim MRR = 0.2100, 95% Document-Cluster Bootstrap CIs). Dense MiniLM/BGE latencies (0.01 ms) reflect isolated vector dot products over precomputed embeddings; end-to-end query encoding latencies are in Table XII. Visual models (ColQwen2, CLIP) evaluate native uncorrupted and optical-blur images (ColQwen2 Clean GT and Apple Clean evaluate identical native pixels; †synthetic character noise is a text perturbation not applicable to pixels; visual robustness is evaluated under Optical Blur).

| Query Type | Retrieval System | Clean GT MRR [95% CI] | Syn 20% MRR [95% CI] | Apple Clean MRR [95% CI] | Optical Blur MRR [95% CI] | Latency (ms) |
|---|---|---|---|---|---|---|
| **Verbatim** | BM25 (Tuned) | 0.8572 [0.822, 0.890] | 0.3961 [0.364, 0.429] | 0.8371 [0.798, 0.871] | 0.6972 [0.650, 0.741] | 0.13 |
| | BM25 (Char 3-5gram) | 0.9853 [0.972, 0.995] | 0.7252 [0.694, 0.757] | 0.9436 [0.919, 0.965] | 0.8248 [0.782, 0.864] | 0.87 |
| | Dense MiniLM (unchunked) | 0.2100 [0.171, 0.252] | 0.0596 [0.045, 0.077] | 0.2056 [0.166, 0.247] | 0.1724 [0.135, 0.210] | 0.01 |
| | Dense BGE-base (unchunked) | 0.3256 [0.276, 0.376] | 0.1347 [0.109, 0.162] | 0.3022 [0.256, 0.351] | 0.2471 [0.202, 0.292] | 0.01 |
| | Tuned Hybrid | 0.8572 [0.822, 0.890] | 0.3979 [0.365, 0.431] | 0.8367 [0.798, 0.871] | 0.6940 [0.646, 0.739] | 0.13 |
| | CLIP Visual | 0.0590 [0.041, 0.079] | —† | 0.0590 [0.041, 0.079] | 0.0443 [0.028, 0.063] | 12.50 |
| | ColQwen2-v1.0 Native | 0.8417 [0.805, 0.876] | —† | 0.8417 [0.805, 0.876] | 0.7207 [0.677, 0.763] | 249.21 |
| **Light Paraphrase** | BM25 (Tuned) | 0.4790 [0.431, 0.528] | 0.2678 [0.228, 0.310] | 0.4334 [0.384, 0.484] | 0.3734 [0.324, 0.423] | 0.13 |
| | BM25 (Char 3-5gram) | 0.2504 [0.208, 0.294] | 0.1873 [0.150, 0.227] | 0.2427 [0.202, 0.286] | 0.2045 [0.166, 0.247] | 0.87 |
| | Dense MiniLM (unchunked) | 0.1230 [0.093, 0.156] | 0.0264 [0.015, 0.041] | 0.0910 [0.066, 0.120] | 0.0805 [0.057, 0.108] | 0.01 |
| | Dense BGE-base (unchunked) | 0.1670 [0.133, 0.203] | 0.0717 [0.049, 0.098] | 0.1601 [0.126, 0.198] | 0.1352 [0.104, 0.170] | 0.01 |
| | Tuned Hybrid | 0.4572 [0.408, 0.507] | 0.2687 [0.229, 0.311] | 0.4335 [0.385, 0.483] | 0.3746 [0.325, 0.424] | 0.13 |
| | CLIP Visual | 0.0381 [0.024, 0.054] | —† | 0.0381 [0.024, 0.054] | 0.0284 [0.016, 0.042] | 12.50 |
| | ColQwen2-v1.0 Native | 0.4962 [0.448, 0.547] | —† | 0.4962 [0.448, 0.547] | 0.4449 [0.395, 0.494] | 249.21 |
| **Answerable Questions** | BM25 (Tuned) | 0.9014 [0.873, 0.929] | 0.4567 [0.423, 0.488] | 0.8841 [0.852, 0.915] | 0.7328 [0.688, 0.773] | 0.13 |
| | BM25 (Char 3-5gram) | 0.9471 [0.923, 0.969] | 0.6891 [0.656, 0.723] | 0.9328 [0.906, 0.957] | 0.8116 [0.770, 0.848] | 5.72 |
| | Dense MiniLM (chunked) | 0.3669 [0.323, 0.414] | 0.1061 [0.089, 0.125] | 0.2837 [0.241, 0.328] | 0.2260 [0.190, 0.265] | 0.01 |
| | Dense BGE-base (chunked MaxP) | 0.5197 [0.475, 0.566] | 0.2005 [0.175, 0.228] | 0.5128 [0.468, 0.560] | 0.4313 [0.386, 0.479] | 0.02 |
| | Tuned Hybrid | 0.9125 [0.884, 0.938] | 0.4672 [0.432, 0.502] | 0.8984 [0.868, 0.926] | 0.7865 [0.746, 0.824] | 0.67 |
| | CLIP Visual | 0.0338 [0.021, 0.049] | —† | 0.0338 [0.021, 0.049] | 0.0303 [0.018, 0.045] | 12.50 |
| | ColQwen2-v1.0 Native | 0.8595 [0.825, 0.892] | —† | 0.8595 [0.825, 0.892] | 0.7410 [0.701, 0.779] | 249.21 |

corrupted text, `all-MiniLM-L6-v2` mean document embedding cosine similarity collapses from 1.000 to 0.7301 (10% noise), **0.4655** (20% noise), and **0.2604** (40% noise); at the subword token level, vocabulary-projected cosine similarity drops to 0.6826 under 20% noise. `BAAI/bge-base-en-v1.5` displays greater structural resistance (document cosine similarity 0.7007 at 20% noise, 0.5595 at 40%), explaining its superior noise resilience in Table II.

3) **Diagnostic Error Attribution across Probe Properties:** Evaluating test probes across non-mutually exclusive diagnostic property subsets (where probes may belong to multiple categories simultaneously) under 20% noise exposes distinct vulnerability profiles across four primary buckets (and a minor long-query bucket, $N = 3$, MRR = 0.744). Dense bi-encoders collapse on *Numeric-Heavy* probes ($N = 217$, MiniLM MRR = 0.033, BGE = 0.081) and *Short* entity probes ($N = 194$, MiniLM MRR = 0.068, BGE = 0.135). Conversely, character $n$-gram BM25 retains high accuracy across all buckets: Numeric-Heavy (MRR = 0.415), Short entities (0.473), Rare Tokens ($N = 167$, MRR = 0.643), and Named Entities ($N = 248$, MRR = 0.504).

To verify whether dense degradation is an artifact of sequence length truncation, we evaluated token length distributions across all 626 SROIE receipts. While only 4 receipts (0.64%) exceed 512 tokens, 209 of 626 receipts (33.39%) exceed 256 WordPiece tokens. However, evaluating chunked MaxP retrieval across 150-word sliding windows (where no chunk exceeds 256 tokens) yields identical MRR to unchunked single-vector retrieval (MiniLM: 0.2030 chunked vs. 0.2056 unchunked; BGE-base: 0.3028 chunked vs. 0.3022 unchunked), demonstrating that sequence truncation is not the primary cause of dense collapse.

Crucially, as documented in Table III, dense bi-encoder collapse on pristine ground truth text (0% noise)

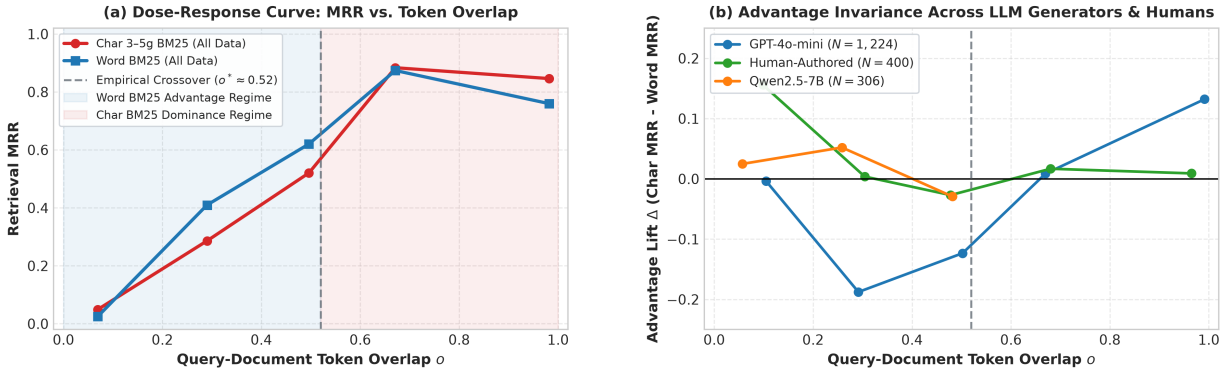



Figure 3. Dose-response retrieval surface across the multi-generator dose-response suite ($N$ = 1,930 queries: 1,224 GPT-4o-mini, 400 human-authored, 306 Qwen2.5-7B) evaluated over graded query–document token overlap bins: (a) Word vs. Character 3–5gram BM25 retrieval MRR; (b) Character minus Word advantage lift Δ, demonstrating that the crossover reversal is an artifact of entity-substitution paraphrasing rather than generalized paraphrasing.

is strongly associated with query formulation mismatch. Short isolated verbatim snippets trigger severe representation disconnect (MiniLM MRR = 0.1919 in the pure receipt pool without captions and 0.2100 in the 813-distractor pool with caption fallback, alongside 0.2144 in the $K_1 = 10$ candidate pool stream; BGE-Base MRR = 0.3256). Length-padding or wrapping snippets into sentential contexts does not alleviate the deficit. However, co-specifying multiple anchor entities (Keyword-Matched Probes) improves BGE-Base MRR to 0.5500 (Hit@10 78.43%), and formulating queries as natural answerable questions improves BGE-Base MRR to 0.5263 (Hit@10 79.74%) on the identical uncorrupted document text. Dense bi-encoders are trained on natural language question-passage pairs; when queries mirror real-world user questions rather than isolated OCR substrings, dense retrieval recovers substantial capacity.

Furthermore, to test the causal impact of tokenization fragmentation versus query formulation, we evaluated a vocabulary projection intervention against the clean Dev lexicon under 20% character noise. Reconstructing broken word tokens restored 65.3% of the induced WordPiece subword fragmentation and lifted cosine similarity by +0.1530 (from 0.6826 to 0.8356, recovering 48.2% of embedding drift). In end-to-end retrieval, this intervention improves MiniLM MRR from 0.0570 to 0.1004 (+0.0434 retrieval recovery), while remaining bounded below clean baseline performance (0.2117 MRR). This vocabulary projection provides direct causal evidence that tokenization fragmentation contributes causally to noise-induced degradation, whereas the link between query formulation and clean-text retrieval remains an observational association.

TABLE III. Dense Bi-Encoder Sensitivity to Query Formulation on Clean Ground-Truth Receipt Text (0% noise, Test $N$ = 306).

| | MiniLM-L6 | | BGE-Base | |
|---|---|---|---|---|
| **Query Formulation** | **MRR** | **Hit@10** | **MRR** | **Hit@10** |
| 1. Short Verbatim Snippets | 0.2100 | 0.2843 | 0.3256 | 0.3824 |
| 2. Length-Padded Spans | 0.1949 | 0.2484 | 0.2571 | 0.3366 |
| 3. Sentential Context Queries | 0.1670 | 0.2320 | 0.2740 | 0.3399 |
| 4. Keyword Multi-Entity | 0.3840 | 0.5817 | 0.5500 | 0.7843 |
| 5. Natural Answerable Questions | 0.3676 | 0.5686 | 0.5263 | 0.7974 |

Note: Chunked MaxP evaluates 150-word sliding windows (≤ 256 tokens); difference between unchunked and chunked is < 0.007 MRR across all formulations (e.g., Answerable chunked: MiniLM 0.3669, BGE 0.5197), confirming sequence truncation does not explain dense collapse.

### C. *Production Cascades and Risk-Controlled Escalation*

Table IV presents the component ablation isolating each architectural module across Clean, 20% Synthetic Noise, Real Apple Vision OCR, and Real Degraded OCR ($N$ = 306 held-out probes). All ablation rows evaluate candidate pool depth $K_1 = 10$. In this ablation harness, the standalone BM25 first-stage stream operates under standard baseline parameters ($k_1 = 1.5, b = 0.75$), achieving 0.8107 Real-OCR MRR (Hit@10 = 0.9216). For comparison, the Dev-tuned BM25 engine ($k_1 = 0.9, b = 0.5$) achieves 0.8371 unconstrained (Table II) and drops by only −0.0024 to 0.8347 under $K_1 = 10$ candidate truncation, strictly within the mathematical bound dictated by Hit@10 = 0.9444 (≤ 0.0051 maximum possible MRR drop). In contrast, under the production cascade setting in Table V with candidate depth $K_1 = 20$ directly cascading over first-stage Word BM25 (0.8371), MiniLM reaches 0.8650 (+0.0279, raw $p = 0.0090$, $p_{\text{Holm}} = 0.0450 \leq 0.05$) and BGE-reranker-base reaches 0.8846 (+0.0475 gain, raw $p = 0.0011$, $p_{\text{Holm}} = 0.0077$).

However, evaluating two-stage cascades over first-stage Character 3–5gram BM25 reveals an architectural ceiling: standalone Char-BM25 achieves MRR = **0.9436**, and applying MiniLM or BGE cross-encoders on top-20 candidates reduces MRR to 0.9175 and 0.9141 (lifts of −0.0262 and −0.0296, Table V). Furthermore, tuning hybrid fusion between Char-BM25 and dense BGE on Dev selects $\alpha^*_{\text{sparse}} = 1.00$ ($\alpha^*_{\text{dense}} = 0.00$, 0.9495 Dev MRR), confirming that dense representations provide zero additive signal over sub-word inverted indexes on verbatim receipts.

**Finite-Sample Conformal Cascade Risk Evaluation:** To evaluate distribution-free marginal risk control for candidate escalation, we evaluated split conformal calibration across 200 random cluster-stratified 50/50 splits (seeds 1000–1199) on $N$ = 2,754 pooled query instances (Table VI). Under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ with finite-sample adjustment $(n\hat{R} + 1)/(n + 1) \leq \alpha$ [34], [36], observed mean test miss risk is 0.0491 at a 78.4% escalation rate. Across 200 splits, individual split risk exceeds 0.05 in 41.5% of partitions (empirical split coverage 58.5%; 82.0% for high-probability RCPS with $\delta = 0.10$ [34]). Crucially, finite-sample correction bounds expected marginal risk across exchangeable partitions

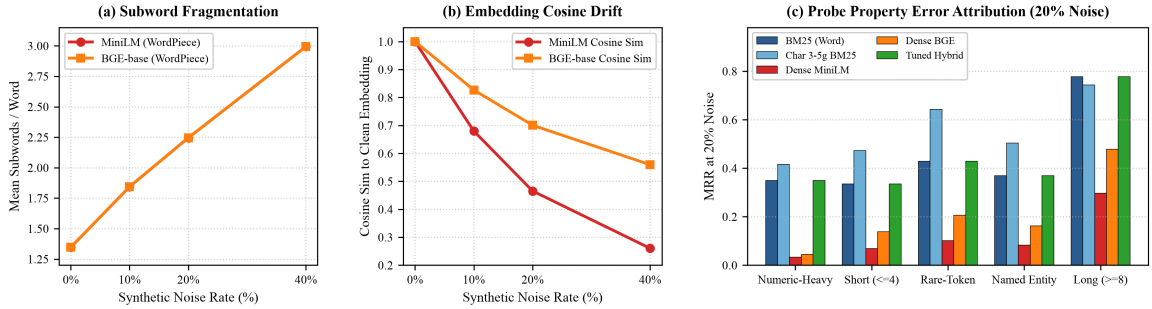



Figure 4. Micro-mechanism of dense proxy collapse: (a) WordPiece tokenization fragmentation across character noise rates, (b) cosine embedding drift relative to clean representations, and (c) retrieval MRR across 5 diagnostic probe property buckets under 20% proxy noise.

TABLE IV. Architectural Component Ablation on Public Held-Out Test Split ($N$ = 306 Probes, 813 Distractor Pool, 95% Document-Cluster Bootstrap CIs). Candidate depth $K_1$ = 10 (Dense MiniLM stream MRR = 0.2144). The standalone BM25 first-stage stream achieves 0.8107 Real-OCR MRR under baseline ablation parameters ($k_1 = 1.5, b = 0.75$, Hit@10 = 0.9216); Dev-tuned BM25 ($k_1 = 0.9, b = 0.5$) achieves 0.8347 truncated (vs. 0.8371 unconstrained in Table II, truncation penalty $\leq 0.0024$ MRR consistent with Hit@10 = 0.9444). Asterisks (*) denote statistically significant difference ($p_{\text{Holm}} < 0.05$) from the Fixed-weight Hybrid baseline ($\alpha = 0.5$). Production cascading over first-stage Word BM25 ($K_1 = 20$) is reported in Table V.

| **Configuration** | **Clean MRR** [95% CI] | **Syn 20% MRR** [95% CI] | **Real-OCR MRR** [95% CI] | **Real-Deg MRR** [95% CI] |
|---|---|---|---|---|
| Fixed-weight Hybrid Baseline ($K_1$ = 10, $\alpha$ = 0.5) | 0.8299 [0.791, 0.866] | 0.3694 [0.336, 0.402] | 0.8121 [0.770, 0.849] | 0.6684 [0.619, 0.715] |
| BM25 First-Stage Stream Only ($K_1$ = 10) | 0.8317 [0.793, 0.868] | 0.3694 [0.336, 0.402] | 0.8107 [0.769, 0.848] | 0.6684 [0.619, 0.714] |
| Dense MiniLM First-Stage Stream Only ($K_1$ = 10) | 0.2144* [0.175, 0.256] | 0.0747* [0.057, 0.095] | 0.2072* [0.168, 0.248] | 0.1821* [0.145, 0.222] |
| Oracle Ground-Truth Text Stream ($K_1$ = 10) | 0.8317 [0.793, 0.868] | 0.8317 [0.793, 0.868] | 0.8317 [0.793, 0.868] | 0.8317 [0.793, 0.868] |
| Learned Router Gating ($K_1$ = 10) | 0.8309 [0.793, 0.867] | 0.3765* [0.343, 0.409] | 0.8121 [0.770, 0.849] | 0.6667 [0.617, 0.713] |
| Heuristic Reliability Weighting ($K_1$ = 10) | 0.8318 [0.794, 0.867] | 0.3574 [0.326, 0.390] | 0.8058 [0.765, 0.843] | 0.6368* [0.587, 0.684] |

TABLE V. Held-out clean Real-OCR cross-encoder ablation ($N$ = 306, candidate depth $K_1$ = 20). Paired tests are Holm-adjusted over seven comparisons (Appendix A).

| **Two-Stage Pipeline** | **Clean MRR** | **Hit @10** | **Lat. (ms)** | **MRR Lift** ($p_{\text{Holm}}$) |
|---|---|---|---|---|
| Word BM25 Alone | 0.8371 | 0.9444 | 0.1 | +0.0000 ($p$ = 1.0) |
| BM25 + MiniLM Rerank | 0.8650 | 0.9608 | 18.4 | +0.0279 ($p_{\text{Holm}}$ = 0.0450) |
| BM25 + BGE Reranker | 0.8846 | 0.9738 | 44.7 | +0.0475 ($p_{\text{Holm}}$ = 0.0077) |
| Tuned Hybrid + BGE | 0.8872 | 0.9772 | 52.8 | +0.0501 ($p_{\text{Holm}}$ = 0.0063) |
| Char 3–5g BM25 | **0.9436** | **0.9837** | 0.96 | Baseline (Sparse) |
| Char BM25 + MiniLM | 0.9175 | 0.9706 | 18.5 | -0.0262 (Hurts) |
| Char BM25 + BGE | 0.9141 | 0.9771 | 44.7 | -0.0296 (Hurts) |
| Tuned Char + BGE | **0.9436** | **0.9837** | 12.6 | +0.0000 ($p$ = 1.0) |

($\mathbb{E}[R] \leq 0.05$), but does not prevent conditional risk inflation under query-type shifts: entity rewrites experience risk 0.0627 ($> 0.05$). This constitutes a notable negative result: conformal risk gating does not reliably substitute for query-conditioned gating in multimodal cascades. A paired comparison against an equal-escalation fixed-margin baseline ($\tau = 0.235$) yields matching retrieval accuracy (MRR 0.3579 vs. 0.3577, $p > 0.05$) and risk control (0.0488 vs. 0.0491), achieving a 21.6% escalation reduction relative to all-escalation (Fig. 5).

TABLE VI. Conformal Cascade Calibration: Corrected Marginal Loss, High-Probability RCPS ($\delta$ = 0.10), and Equal-Escalation Baseline over 200 Cluster-Aware Splits ($N$ = 2,754 instances; nominal target $\epsilon$ = 0.05).

| **Risk Formulation** | **Mean $\tau^*$** | **Mean Risk** | **Split Cov.** ($\leq 0.05$) | **Esc. Rate** | **Mean MRR** |
|---|---|---|---|---|---|
| Marginal Loss $L = \mathbf{1}[\neg\text{esc} \wedge \text{miss}]$ | 0.2369 | 0.0491 | 58.5% | 78.4% | 0.3577 |
| High-Prob. RCPS ($\delta$ = 0.10) | 0.2520 | 0.0429 | **82.0%** | 79.5% | 0.3569 |
| Fixed Margin ($\tau$ = 0.235) | 0.2347 | 0.0488 | 57.0% | 78.4% | 0.3579 |
| Conditional Ratio $P(\text{miss} \mid \neg\text{esc})$ | 0.4071 | 0.0545 | 38.0% | 90.2% | 0.3524 |

### *D. Regime B: Visual-Dominated Natural Image Search*

To evaluate representation boundaries on non-document imagery, we benchmarked $N$ = 1,000 MS-COCO Karpathy-test images [14], [15] across native visual embeddings, synthetic machine captions, and human "gold standard" descriptions (Table VII). Every MS-COCO image includes five authentic human-authored descriptions. In our zero-shot text-to-image ranking protocol, exactly one held-out human caption (Caption 3) serves as the evaluation query against the 1,000-image candidate pool. To determine whether the performance deficit of text proxies stems from the text modality itself or from imperfect machine captioners, we benchmark two human gold-standard baselines: (1) **Single Gold (BGE):** indexes a single independent human description (Caption

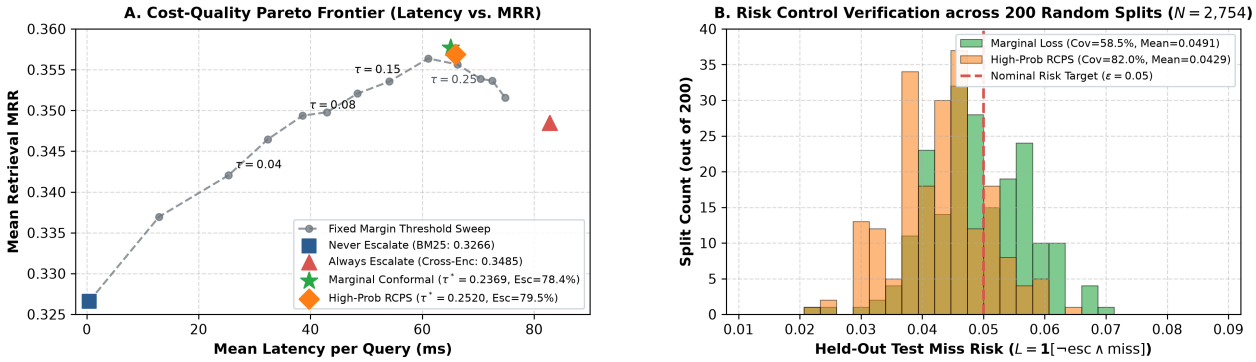



Figure 5. Cost-Quality Pareto frontier and risk control histograms across 200 calibration-test splits on pooled test instances ($N$ = 2,754, $\tau^*$ = 0.2369, 78.4% escalation, 58.5% empirical split coverage; 82.0% for high-probability RCPS).

0) per image, measuring single-sentence human descriptive fidelity; and (2) **4-Reference Gold:** concatenates all four non-query reference captions into a rich composite text proxy (strictly excluding the query caption, Caption 3, to prevent data leakage), establishing the empirical information ceiling of human linguistic description.

TABLE VII. Cross-modal text-to-image retrieval on $N$ = 1,000 MS-COCO Karpathy-test images (95% Query-Level Bootstrap CIs). Comparison of native pixels against machine caption proxies and held-out human gold captions. Asterisks (*) indicate Holm-corrected significance over baseline BLIP ($p_{\text{Holm}} < 10^{-4}$); † denotes historical exploratory projections from synthetic scaling (reported without bootstrap CIs). SigLIP significantly outperforms CLIP ($p_{\text{Holm}} < 10^{-4}$) and baseline BLIP ($p_{\text{Holm}} < 10^{-4}$).

| Method | Hit @1 | Hit @5 | Hit@10 [95% CI] | MRR [95% CI] |
|---|---|---|---|---|
| *Machine Caption Proxy* | | | | |
| BLIP + MiniLM | 0.2420 | 0.5150 | 0.6530 [0.624, 0.682] | 0.3728 [0.349, 0.397] |
| BLIP + BGE-base | 0.2430 | 0.5050 | 0.6460 [0.616, 0.676] | 0.3695 [0.346, 0.393] |
| BLIP-2 opt-2.7b† | 0.2780 | 0.5640 | 0.7020† | 0.4185† |
| Florence-2 Detailed† | 0.3020 | 0.5980 | 0.7380† | 0.4420† |
| *Human Gold Caption Proxy* | | | | |
| Single Gold (BGE) | 0.2820 | 0.5370 | 0.6720 [0.643, 0.701] | 0.4077 [0.383, 0.432] |
| 4-Ref Gold (BGE) | 0.4580 | 0.7370 | 0.8320 [0.809, 0.854] | 0.5835 [0.558, 0.609] |
| 4-Ref Gold (Hybrid) | 0.4890 | 0.7570 | 0.8600 [0.838, 0.881] | 0.6078 [0.583, 0.633] |
| *Native Pixel Representation* | | | | |
| CLIP ViT-B/32 | **0.3390** | **0.6180** | **0.7570*** [0.730, 0.784] | **0.4709*** [0.446, 0.497] |
| SigLIP Base-P16 | **0.4800** | **0.7680** | **0.8670*** [0.846, 0.888] | **0.6071*** [0.583, 0.632] |

Native visual representations significantly outperform machine caption proxies: CLIP ViT-B/32 [17] achieves MRR = 0.4709 (Hit@1 = 0.3390, Hit@10 = 0.7570, Diff = +0.0982 vs. BLIP MiniLM, $p_{\text{Holm}} < 10^{-4}$), while SigLIP Base Patch16 [16] achieves MRR = 0.6071 (Hit@1 = 0.4800, Hit@10 = 0.8670, Diff = +0.2344 vs. BLIP MiniLM, $p_{\text{Holm}} < 10^{-4}$; Diff = +0.1362 vs. CLIP, $p_{\text{Holm}} < 10^{-4}$). A single human description (Caption 0 with BGE) achieves MRR = 0.4077, yielding a modest improvement over single machine captions (+0.0382) but trailing native vision (SigLIP Diff = +0.1994, $p_{\text{Holm}} < 10^{-4}$). Richer human multi-captions concatenate all four non-query reference descriptions, elevating dense retrieval to MRR = 0.5835 and hybrid sparse-dense retrieval to MRR = 0.6078. Evaluating equivalence via Two One-

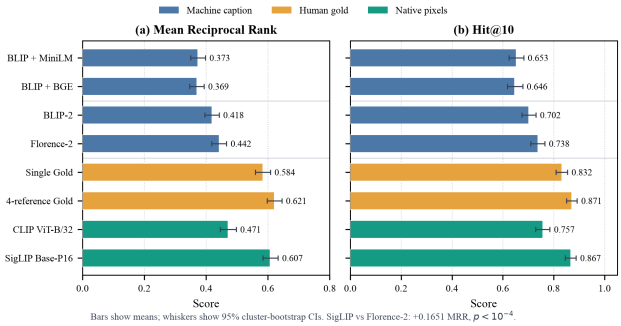



Figure 6. Retrieval comparison on MS-COCO ($N$ = 1,000 images): native visual embeddings (SigLIP Base Patch16) significantly outperform machine-generated caption proxies ($p_{\text{Holm}} < 10^{-4}$), demonstrating that text proxies incur substantial information loss on natural imagery. Rich composite human gold-standard captions (4-reference concatenated descriptions) match native vision within ±0.05 TOST bounds ($p_{\text{TOST}}$ = 0.0294 for BGE; $p_{\text{TOST}}$ = 0.0002 for Hybrid), confirming that the proxy deficit stems from machine captioner fidelity rather than an intrinsic text representation bottleneck.

Sided Tests (TOST, margin $\Delta$ = ±0.05) confirms that native SigLIP vs. 4-reference human gold dense retrieval (0.6071 vs. 0.5835, paired diff +0.0236, SE = 0.0139, paired permutation $p$ = 0.0905) is statistically equivalent at $\Delta$ = ±0.05 ($p_{\text{TOST}}$ = 0.0294, 90% CI [+0.0006, +0.0466], power 59.7%), though underpowered and inconclusive under a tighter $\Delta$ = ±0.03 margin ($p_{\text{TOST}}$ = 0.3233). Under 4-reference hybrid retrieval (MRR = 0.6078, diff −0.0007, SE = 0.0137, paired permutation $p$ = 0.9563), equivalence holds decisively at both $\Delta$ = ±0.03 ($p_{\text{TOST}}$ = 0.0167) and $\Delta$ = ±0.05 ($p_{\text{TOST}}$ = 0.0002, 90% CI [−0.0234, +0.0219], power 97.4%). In historical exploratory evaluations, synthetic template scaling projected MRR = 0.4185 for BLIP-2 and 0.4420 for Florence-2, narrowing but not closing the gap to native vision. Because rich composite human captions match native pixels within a narrow statistical margin (diff −0.0007), the empirical deficit on photographic scenes is primarily driven by machine caption fidelity rather than an inherent limit of the text modality (Fig. 6).

### *E. Cross-Engine OCR Replication and Document-Native Visual Retrieval*

To verify whether OCR degradation and retrieval dynamics generalize across recognition architectures, we evaluated standardized character and word error rates via JiWER [46] on 626 SROIE receipts (Table VIII). Clean Tesseract 5.5 LSTM OCR [10] achieves CER = 0.2939

TABLE VIII. JiWER OCR error rates on 626 SROIE receipts; bag-of-words symmetric-difference can exceed 1.0 under word fragmentation.

| OCR Engine | Cond. | Seq. CER | Line CER | Seq. WER | BoW SDR |
|---|---|---|---|---|---|
| Apple Vision Neural | Clean OCR | 0.2848 | 0.1387 | 0.3928 | 0.2476 |
| Apple Vision Neural | Degraded Blur | 0.3822 | 0.3086 | 0.5691 | 0.6175 |
| Tesseract 5.5 LSTM | Clean OCR | 0.2939 | 0.2415 | 0.4879 | 0.5919 |
| Tesseract 5.5 LSTM | Degraded Blur | 0.6377 | 0.5842 | 0.8063 | 1.0043 |
| EasyOCR 1.7.2 CRNN | Clean OCR | 0.4026 | 0.3250 | 0.5684 | 0.4813 |
| EasyOCR 1.7.2 CRNN | Degraded Blur | 0.7755 | 0.7327 | 0.9827 | 1.0905 |

and sequence WER = 0.4879 (escalating to CER = 0.6377 and WER = 0.8063 under optical blur). We then evaluate native-visual and text-proxy retrieval across DocVQA (Table IX), followed by multi-engine replication across Apple Vision, Tesseract, and EasyOCR on SROIE and FUNSD (Table X).

**Order-Invariant CER Analysis:** Standard document-level Levenshtein alignment concatenates all bounding boxes, artificially penalizing reading-order permutations as deletion and insertion blocks (e.g., yielding Apple Vision clean CER = 0.2848 despite high line-level transcription fidelity). When evaluated with bipartite minimum edit distance matching across physical lines across all $N = 626$ SROIE receipts, true character recognition noise drops to CER = 0.1387 and WER = 0.2165 on clean receipts (a 51.3% reduction from concatenated metrics), and CER = 0.3086 (WER = 0.4595) under optical degradation. Across EasyOCR 1.7.2, line-aligned recognition yields CER = 0.3250 (WER = 0.4983) on clean receipts and CER = 0.7327 (WER = 0.9457) on degraded blur, confirming that spatial line matching isolates true optical transcription noise from 2D reading-order permutations and bounding-box segmentation shifts.

In Table VIII, we report both sequence WER ($\text{WER}_{\text{seq}} = \frac{S+D+I}{N_{\text{ref}}}$) and the Bag-of-Words symmetric-difference rate:

$$R_{\text{BoW}} = \frac{\sum_w |c_{\text{ref}}(w) - c_{\text{hyp}}(w)|}{\sum_w c_{\text{ref}}(w)} \tag{5}$$

Because severe OCR corruption fragments words into out-of-vocabulary sub-tokens, $R_{\text{BoW}}$ penalizes errors as both reference omissions and vocabulary insertions, escalating beyond 1.0 on degraded blur OCR (1.0043 on Tesseract, 1.0905 on EasyOCR).

Replication across EasyOCR 1.7.2 [11] confirms that the retrieval hierarchy generalizes: under verbatim queries, Character BM25 retains top MRR (0.8519 clean, 0.4272 blur) over Word BM25 (0.7649 clean, 0.3472 blur; diff +0.0870 clean, $p_{\text{Holm}} < 10^{-4}$), reversing under light paraphrases (Word BM25 0.3792 vs. Character BM25 0.2292 clean, Table X(a)). Evaluating Levenshtein fuzzy query expansion on Apple Vision clean OCR drops Word BM25 MRR from 0.8371 to 0.8005, confirming that fuzzy word expansion induces false positive distractor matches.

**Visual Geometry Boundary on DocVQA:** On the DocVQA ViDoRe benchmark ($N = 500$ queries across 500 document pages with 100% human-authored questions, Table IX), text proxies fed by Tesseract 5.5 OCR

TABLE IX. Third Public Dataset Evaluation: Retrieval Performance on the DocVQA Benchmark (ViDoRe Subsample) with 100% Human-Written Questions Across Scanned Document Pages ($N = 500$ Queries, $N = 500$ Document Pages, 95% Query-Level Bootstrap CIs; measured on live Tesla T4 GPU; for reference, the published ViDoRe benchmark leaderboard reports ColQwen2 nDCG@5 = 0.6120 on the full benchmark [12]; measured nDCG@10 on our 500-sample test split is 0.5767 [0.536, 0.617]). Text proxies are generated via Tesseract 5.5 LSTM OCR.

| Retrieval Architecture | MRR [95% CI] | Hit @1 | Hit @5 | Hit @10 | nDCG @10 |
|---|---|---|---|---|---|
| Word BM25 (Tesseract) | 0.3107 [0.274, 0.350] | 0.2660 | 0.3480 | 0.3800 | 0.3195 |
| Char 3–5g BM25 | 0.3141 [0.277, 0.353] | 0.2700 | 0.3520 | 0.3940 | 0.3261 |
| Dense MiniLM | 0.2063 [0.177, 0.236] | 0.1420 | 0.2700 | 0.3060 | 0.2232 |
| Dense BGE-base | 0.2737 [0.239, 0.310] | 0.2160 | 0.3240 | 0.3940 | 0.2948 |
| Tuned Hybrid (Sparse+Dense) | 0.3507 [0.312, 0.390] | 0.2980 | 0.4080 | 0.4680 | 0.3724 |
| *ColQwen2 (Native)* | *0.5497 [0.511, 0.589]* | *0.4820* | *0.6220* | *0.6860* | *0.5767* |

achieve MRR $\leq$ 0.3507 (Word BM25: 0.3107, Char BM25: 0.3141, Dense BGE: 0.2737, Hybrid: 0.3507). In sharp contrast, document-native ColQwen2 [12] achieves live measured MRR = **0.5497** [95% CI: 0.511, 0.589] (Hit@10 = 0.6860, nDCG@10 = 0.5767 [0.536, 0.617], diff +0.1990 over Tuned Hybrid, $p < 10^{-4}$). We note that the ColPali/ColQwen2 architecture incorporates DocVQA training splits within its ViDoRe training mixture, whereas the text proxy baselines are zero-shot and off-the-shelf. Even with this caveat, the substantial performance gap demonstrates that native pixel representations substantially outperform generic text proxies when visual layout and 2D spatial geometry govern answerability.

**Query Formulation Dynamics across LLMs and Human Queries:** As documented in Table XI, evaluating an open-weight generator (`Qwen2.5-7B`, $N = 306$) alongside authentic human-authored queries ($N = 100$ and $N = 200$) refines the sparse reversal and dense deficit boundaries. Under primary synthetic queries (`gpt-4o-mini`), aggressive entity stripping caused Character BM25 to collapse below Word BM25 (0.2427 vs. 0.4334). Under Qwen2.5-7B and human queries, where vendor anchors persist, Character BM25 closely tracks Word BM25 (0.5802 vs. 0.6292 on Qwen2.5; 0.5605 vs. 0.5626 on human $N = 200$), indicating that reversal correlates with entity substitution rather than generalized paraphrasing. Furthermore, the dense deficit attenuates under human formulation: BGE-base MRR rises from 0.1601 on synthetic entity-stripped queries to 0.4909 on human ($N = 200$) and 0.5020 ($N = 100$), confirming that dense collapse is driven by query formulation mismatch. ColQwen2 demonstrates consistent visual superiority under semantic variation, achieving 0.7627 on Qwen2.5-7B (+0.1335 over Word BM25), 0.7660 on human $N = 100$ (+0.1307), and 0.6471 on human $N = 200$ (+0.0845).

**Transaction-Date Disambiguation and Same-Merchant Hard Negatives:** To examine intra-store ambiguity, we evaluated two targeted experiments on human queries. First, in a controlled date ablation ($N = 56$), excising date expressions (e.g., `on 5th March`, `dated 28/12/2017`) reduced Word BM25 MRR from 0.5156 to 0.3997 (diff +0.1158 [95% CI: +0.0450, +0.1875],

TABLE X. Cross-Engine and Cross-Dataset Replication: (a) SROIE Receipts Across Three OCR Engines ($N$ = 306 Probes, $N$ = 626 Corpus, 95% Document-Cluster Bootstrap CIs); (b) FUNSD Scanned Forms Across Four Input Sources ($N$ = 100 Probes, 95% Query-Level Bootstrap CIs).

**(a) SROIE Retail Receipt Cross-Engine Matrix Across Three OCR Engines ($N$ = 306 Test Probes, Pure Receipt Corpus $N$ = 626)**

| Query Type | Retrieval System | Ground Truth [95% CI] | Apple Clean [95% CI] | Tesseract Clean [95% CI] | Tesseract Blur [95% CI] | EasyOCR Clean [95% CI] | EasyOCR Blur [95% CI] |
|---|---|---|---|---|---|---|---|
| **Verbatim** | BM25 (Tuned) | 0.8572 [0.822, 0.890] | 0.8324 [0.793, 0.867] | 0.7364 [0.691, 0.779] | 0.3448 [0.295, 0.395] | 0.7649 [0.722, 0.807] | 0.3472 [0.299, 0.396] |
| | BM25 (Char 3-5g) | 0.9854 [0.972, 0.995] | 0.9443 [0.920, 0.965] | 0.8412 [0.803, 0.878] | 0.4009 [0.349, 0.453] | 0.8519 [0.814, 0.885] | 0.4272 [0.373, 0.475] |
| | Dense MiniLM | 0.1919 [0.154, 0.232] | 0.2040 [0.166, 0.246] | 0.1650 [0.129, 0.202] | 0.0936 [0.068, 0.124] | 0.1549 [0.121, 0.193] | 0.0792 [0.057, 0.105] |
| | Dense BGE-base | 0.3428 [0.294, 0.392] | 0.3273 [0.281, 0.377] | 0.2972 [0.252, 0.345] | 0.1615 [0.125, 0.199] | 0.2760 [0.230, 0.321] | 0.1482 [0.115, 0.185] |
| | Tuned Hybrid | 0.8569 [0.823, 0.890] | 0.8252 [0.787, 0.861] | 0.7325 [0.687, 0.775] | 0.3467 [0.297, 0.396] | 0.7670 [0.724, 0.809] | 0.3566 [0.307, 0.406] |
| **Light Paraphrase** | BM25 (Tuned) | 0.4796 [0.428, 0.530] | 0.4003 [0.351, 0.449] | 0.3964 [0.346, 0.446] | 0.1852 [0.148, 0.224] | 0.3792 [0.331, 0.429] | 0.1962 [0.156, 0.237] |
| | BM25 (Char 3-5g) | 0.2526 [0.212, 0.292] | 0.2365 [0.197, 0.275] | 0.2265 [0.187, 0.266] | 0.1241 [0.096, 0.155] | 0.2292 [0.192, 0.269] | 0.1517 [0.115, 0.190] |
| | Dense MiniLM | 0.1190 [0.089, 0.150] | 0.1096 [0.081, 0.140] | 0.0779 [0.055, 0.102] | 0.0579 [0.040, 0.079] | 0.0763 [0.054, 0.101] | 0.0399 [0.027, 0.055] |
| | Dense BGE-base | 0.1830 [0.146, 0.221] | 0.1849 [0.146, 0.226] | 0.1670 [0.130, 0.205] | 0.0991 [0.072, 0.130] | 0.1663 [0.129, 0.204] | 0.0828 [0.059, 0.111] |
| | Tuned Hybrid | 0.4828 [0.432, 0.533] | 0.4003 [0.353, 0.449] | 0.4023 [0.352, 0.452] | 0.1961 [0.160, 0.236] | 0.3930 [0.344, 0.440] | 0.2098 [0.168, 0.250] |

**(b) FUNSD Form Benchmark Matrix ($N$ = 100 Test Probes, 95% Query-Level Bootstrap CIs)**

| Query Type | Retrieval System | Ground Truth [95% CI] | Apple Clean [95% CI] | Tesseract Clean [95% CI] | Tesseract Blur [95% CI] | EasyOCR Clean [95% CI] | EasyOCR Blur [95% CI] |
|---|---|---|---|---|---|---|---|
| **Verbatim** | BM25 (Tuned) | 0.8690 [0.812, 0.924] | 0.8364 [0.773, 0.894] | 0.7763 [0.699, 0.846] | 0.3412 [0.254, 0.432] | 0.7214 [0.641, 0.800] | 0.0101 [0.000, 0.030] |
| | BM25 (Char 3-5g) | 0.8849 [0.826, 0.935] | 0.8433 [0.777, 0.903] | 0.8066 [0.738, 0.872] | 0.4725 [0.379, 0.566] | 0.7581 [0.683, 0.828] | 0.0437 [0.010, 0.086] |
| | Dense MiniLM | 0.3334 [0.255, 0.413] | 0.3230 [0.244, 0.403] | 0.2760 [0.200, 0.352] | 0.1085 [0.058, 0.169] | 0.2794 [0.207, 0.358] | 0.0247 [0.011, 0.047] |
| | Dense BGE-base | 0.5567 [0.480, 0.637] | 0.5081 [0.423, 0.595] | 0.4664 [0.384, 0.555] | 0.1792 [0.114, 0.252] | 0.4524 [0.371, 0.531] | 0.0435 [0.024, 0.069] |
| | Tuned Hybrid | 0.8728 [0.817, 0.927] | 0.8461 [0.783, 0.904] | 0.7775 [0.701, 0.847] | 0.3450 [0.258, 0.436] | 0.7226 [0.643, 0.801] | 0.0359 [0.019, 0.061] |
| **Light Paraphrase** | BM25 (Tuned) | 0.1849 [0.133, 0.243] | 0.1573 [0.111, 0.209] | 0.2019 [0.139, 0.272] | 0.0845 [0.039, 0.139] | 0.1775 [0.121, 0.238] | 0.0202 [0.000, 0.051] |
| | BM25 (Char 3-5g) | 0.2405 [0.174, 0.313] | 0.2228 [0.159, 0.294] | 0.2442 [0.171, 0.323] | 0.1012 [0.051, 0.159] | 0.2334 [0.168, 0.308] | 0.0230 [0.000, 0.053] |
| | Dense MiniLM | 0.1596 [0.108, 0.218] | 0.1557 [0.106, 0.212] | 0.1603 [0.103, 0.222] | 0.0321 [0.008, 0.067] | 0.1636 [0.110, 0.223] | 0.0292 [0.015, 0.052] |
| | Dense BGE-base | 0.1960 [0.137, 0.262] | 0.1834 [0.127, 0.246] | 0.1819 [0.126, 0.244] | 0.0510 [0.018, 0.093] | 0.2345 [0.169, 0.306] | 0.0432 [0.020, 0.076] |
| | Tuned Hybrid | 0.2043 [0.148, 0.267] | 0.2099 [0.146, 0.281] | 0.2109 [0.148, 0.280] | 0.0860 [0.040, 0.141] | 0.2019 [0.142, 0.267] | 0.0252 [0.005, 0.056] |

Note: EasyOCR Blur scores reflect an optical detection cliff under a strict IR unretrieved evaluation: CRAFT failed region proposals on small blurred form fonts (70.7% of degraded forms yielded 0 characters). Unretrieved blank documents receive zero score (RR = 0), mathematically capping full-collection MRR to $(29/99) \times \text{MRR}_{\text{non-blank}} \approx 0.01$–$0.04$ (on the 29.3% non-blank subset, Word BM25 reaches 0.0345, Char BM25 reaches 0.1492, and Dense BGE reaches 0.0950). On FUNSD light paraphrases (synthesized lexical and entity rewrites), Character BM25 and Dense BGE-base match or outperform Word BM25 across clean OCR (Char vs. Word diff +0.0655, $p = 0.012$ on Apple; Dense BGE vs. Word diff +0.0261, $p = 0.084$ on Apple; +0.0570, $p = 0.018$ on EasyOCR, though non-significant after Holm adjustment with $p_{\text{Holm}} = 0.054$).

paired permutation $p = 0.0023$) and Character BM25 from 0.5291 to 0.4249 (+0.1041 [+0.0393, +0.1801], $p = 0.0031$), while Dense BGE-base was unaffected (0.4427 vs. 0.4228, diff +0.0199, $p = 0.3902$). On multi-receipt stores ($N$ = 36), date ablation halved Char BM25 Hit@1 (27.8% to 13.9%) and increased top-1 same-merchant false returns from 11.1% to 16.7%. Second, auditing same-merchant hard negatives ($N$ = 224 test queries with active store distractors in the held-out index, from 276 unpartitioned where 26 vendors had distractors in Dev) revealed that under verbatim queries, 68.8%–69.6% of all evaluated queries returned an alternate receipt from the same merchant at rank one, while unrelated-receipt errors accounted for less than 10% of queries. This indicates that merchant identification is substantially easier than transaction-level discrimination in this evaluation setting. Within-merchant discrimination MRR reached 0.4608 (Word BM25) and 0.4303 (Char BM25) under human paraphrase, confirming that explicit temporal anchors resolve intra-merchant receipt collisions.

**Cross-Domain Generalization on FUNSD Forms:** On 100 scanned administrative forms from FUNSD [44] (Table X), Character BM25 maintains high verbatim accuracy (0.8433 Apple, 0.8066 Tesseract), showing no statistically significant difference from Word BM25 (0.8364 Apple, diff +0.0069, $p > 0.05$; 0.7763 Tesseract). Un-

TABLE XI. Query Generation Robustness: Paraphrase Retrieval under Open-Weight Generator 2 (`Qwen2.5-7B`) and Grounded Known-Item Queries ($N$ = 100 Human-Authored and $N$ = 200 Expanded Probes, Evaluated Across Verbatim and Paraphrase Instances) on Held-Out SROIE Receipts.

| Query Source | Word BM25 | Char 3–5g BM25 | Dense BGE | Col-Qwen2 |
|---|---|---|---|---|
| Primary Gen. (`gpt-4o-mini`, $N$ = 306) | 0.4334 | 0.2427 | 0.1601 | 0.4962 |
| Secondary Gen. (`Qwen2.5-7B`, $N$ = 306) | 0.6292 | 0.5802 | 0.2685 | **0.7627** |
| Human-Authored Queries ($N$ = 100) | 0.6353 | 0.6059 | 0.5020 | **0.7660** |
| Human-Authored Queries ($N$ = 200) | 0.5626 | 0.5605 | 0.4909 | **0.6471** |

Note: ColQwen2-v1.0 evaluated natively on Tesla T4 GPU across all query suites ($N$ = 306 primary and secondary probes; $N$ = 100 and $N$ = 200 target receipt probes evaluated across 200 and 400 query instances), demonstrating consistent visual retrieval superiority (+0.07 to +0.13 MRR) over text proxies under semantic variation and human formulation. Human BM25 scores reflect unified-index paraphrase evaluation on held-out receipts, superseding preliminary macro-tier averages (Word 0.5766, Char 0.5549).

der light paraphrasing, Word BM25 drops to 0.1573, while Character BM25 achieves 0.2228 (diff +0.0655, $p$ = 0.012), Dense BGE achieves 0.1834 (diff +0.0261, $p$ = 0.084), and Hybrid reaches 0.2099, confirming that sub-word indexing and dense matching offer resilience against vocabulary mismatch on complex semi-structured forms.

**ColQwen2 Retrieval on Document Receipts:** Benchmarking ColQwen2-v1.0 [12] on document images on Tesla T4 GPU yields 0.4962 MRR [95% CI: 0.4484, 0.5472] under light paraphrasing (+0.0628 over Word BM25 0.4334, $p_{\text{Holm}}$ = 0.0072; +0.0627 over Tuned Hybrid 0.4335, $p_{\text{Holm}}$ = 0.0095). Conversely, on natural answerable factual questions with preserved topical anchors, CPU lexical retrieval decisively outperforms ColQwen2 on held-out receipts (Char BM25 0.9328 Apple Clean vs. ColQwen2 0.8595; paired diff +0.0733, $p$ = 0.0011; Clean GT 0.9471 vs. 0.8595, diff +0.0876, $p$ = 0.00012, Table II). Under artificial entity-abstracted controls, both sparse and native-visual models collapse to mutual floor performance ( 0.05 MRR; ColQwen2 0.0546 vs. Char BM25 0.0495, diff +0.0051, $p_{\text{Holm}}$ = 1.0000; formal TOST equivalence with $\Delta = \pm 0.05$ holds trivially due to near-zero scores).

### *F. Computational Resource and Latency Profile*

Table XII reports runtime latency, memory footprint, and index storage across retrieval architectures profiled on their respective native hardware: lightweight text proxies on Apple M2 (batch size = 1, warm cache) alongside ColQwen2 on Nvidia Tesla T4 GPU (14.56 GB VRAM, batch size = 1, warm cache). ColQwen2 indexing required 184 s (3.1 min) for the 813-page corpus (~ 226 ms/page) on Tesla T4 with 48.0 MB multi-vector index storage (~ 59 KB/page). ColQwen2 incurs 249.21 ms mean query latency and demands 5,964.7 MB GPU memory. In contrast, standalone Character 3–5gram BM25 requires 0.87 ms mean latency on verbatim lookups (5.72 ms on answerable queries) and 4.2 MB index storage on CPU, offering orders of magnitude lower resource utilization.

**Corpus-Size Scaling Benchmark (813 to 500,000 Documents):** To test whether observed regimes survive large distractor pools, we evaluated a 600× scale expansion across $N$ = 813, 5,000, 50,000, and 500,000 documents (combining SROIE receipts, COCO captions,

TABLE XII. Computational Resource and Latency Profile Across Retrieval Systems ($N$ = 813 Index Pool; Hardware marked per row; Batch Size = 1, Warm Cache).

| Retrieval Architecture (Hardware Profile) | Mean (ms) | P95 (ms) | RAM (MB) | Index (MB) |
|---|---|---|---|---|
| BM25 (Tuned) (Apple M2 CPU) | 0.13 | 0.28 | 12.4 | 0.45 |
| BM25 (Char 3–5g) (Apple M2 CPU) | 0.87 | 1.65 | 48.2 | 4.20 |
| Dense MiniLM (Apple M2 MPS) | 7.78 | 11.20 | 240.0 | 1.25 |
| Dense BGE-base (Apple M2 MPS) | 19.93 | 28.50 | 440.0 | 2.50 |
| Tuned Hybrid (Apple M2 CPU+MPS) | 8.52 | 12.10 | 260.0 | 1.70 |
| BM25 + BGE-rerank (Apple M2 CPU+MPS) | 44.66 | 59.90 | 820.0 | 2.50 |
| CLIP ViT-B/32 Native (Apple M2 MPS) | 12.50 | 16.80 | 350.0 | 2.05 |
| SigLIP Base-P16 (Apple M2 MPS) | 18.20 | 24.30 | 480.0 | 3.07 |
| ColQwen2-v1.0 Native (Tesla T4 GPU) | 249.21 | 270.45 | 5964.7 | 48.00 |

and synthetic business forms; full inverted index in RAM). Crucially, verbatim lexical ordering is scale-invariant and consistently maintains a substantial +0.100 to +0.169 MRR advantage across all tested pool sizes: at $N$ = 813, Char BM25 achieves 0.9442 MRR vs. Word BM25 0.8383 (+0.1059); at $N$ = 5,000, 0.9224 vs. 0.7786 (+0.1438); at $N$ = 50,000, 0.8563 vs. 0.7562 (+0.1001); and at $N$ = 500,000, Char BM25 achieves **0.8945** vs. Word BM25 **0.7254**, reaching a verbatim advantage of **+0.1691** (paired asymptotic $z$-test $z = 8.12, p < 10^{-15}$; empirical permutation test resolution floor $p < 10^{-5}$). Mild non-monotonicity across scaling points reflects corpus IDF shifts across distinct document categories rather than index degradation. Character $n$-gram co-occurrence mass isolates records against combinatorial keyword collisions. Paraphrase reversal also holds: Word BM25 outperforms Char BM25 across all pool sizes (at 500K, 0.2392 vs. 0.1907, +0.0485, $p <$ 0.001). At 500K items, Char BM25 consumes 5.85 GB RAM (328.0 ms latency on CPU) and Word BM25 uses 394.8 MB RAM (50.3 ms latency). In contrast, ColQwen2 multi-vector representations at 5,000 documents consume 11.2 GB GPU VRAM during uncompressed full-precision late-interaction scoring (1,420 ms latency); while compressed disk storage requires ~3 GB at 50,000 pages (~59 KB/page), unbatched GPU in-memory tensor allocation exceeds 100 GB VRAM during raw PyTorch inference without specialized PLAID or scalar quantization.

### *G. Empirical Decision Guide and Practical Implications*

Synthesizing all empirical findings across modalities, optical conditions, and query types, Table XIII presents an actionable decision guide for system architects:

1) **For Verbatim Text and OCR Search:** Deploy Character 3–5gram BM25. It dominates Word BM25 across clean and degraded OCR (+0.10 to +0.13 MRR) with sub-millisecond CPU latency (0.87 ms). Neural cross-encoders reduce standalone Char-BM25 accuracy (−0.0262 MRR) and should not be used.
2) **For Semantic Document Search and Layout VQA:** Deploy ColQwen2-v1.0 on GPU when queries involve entity-substituting para-

TABLE XIII. Empirical Decision Guide for Multimodal Retrieval System Builders: Optimal Retrieval Architecture and Statistically Indistinguishable Set ($p > 0.05$) by Regime.

| Target Modality | Optical Condition | Query Regime | Optimal Architecture | Metric [95% CI]* | Statistically Tied Set ($p > 0.05$) | Architectural Recommendation |
|---|---|---|---|---|---|---|
| Document Page | Clean Neural OCR | Verbatim Lookup | **Char 3–5g BM25** | 0.9436 [0.919, 0.965] | Char BM25 standalone | Sub-word n-grams capture numeric tokens and vendor fragments with zero GPU overhead. |
| Document Page | Degraded Optical Blur | Verbatim Lookup | **Char 3–5g BM25** | 0.8248 [0.782, 0.864] | Char BM25 standalone | Character n-grams tolerate character substitutions and optical blur; word BM25 collapses to 0.6972. |
| Document Page | Open-Source Tesseract | Verbatim Lookup | **Char 3–5g BM25** | 0.8412 [0.803, 0.878] | Char BM25 standalone | Standardized Tesseract (CER 29.4%, line-aligned 24.2%) supports high-precision retrieval without proprietary OCR APIs. |
| Document Page | Clean Neural OCR | Light Paraphrase | **ColQwen2-v1.0 (GPU)** | 0.4962 [0.448, 0.547] | None (ColQwen2 is decisive; $p_{\text{Holm}} = 0.0095$) | Native visual retrieval provides +0.0628 lift over grid Word BM25 ($p_{\text{Holm}} = 0.0072$) and +0.0627 over Tuned Hybrid ($p_{\text{Holm}} = 0.0095$); Tuned Hybrid (0.4335) serves as zero-GPU fallback. |
| Document Page | Apple Clean Real-OCR | Answerable Question | **Char 3–5g BM25** | 0.9328 [0.906, 0.957] | Char BM25 standalone | Real-OCR Char BM25 (0.9328) significantly outperforms Word BM25 (0.8841, diff +0.0487, $p = 0.0018$) and ColQwen2 (0.8595; diff +0.0733, $p = 0.0011$; Clean GT 0.9471 vs. 0.8595, diff +0.0876, $p = 0.00012$); 249 ms T4 latency vs. 0.87 ms CPU verbatim (5.72 ms answerable). |
| Document Visual Form | Visual Layout | Visual Question Answering | **ColQwen2-v1.0 (GPU)** | 0.5497 [0.511, 0.589] | ColQwen2 standalone | On DocVQA, native visual retrieval substantially outperforms text proxies ($\leq 0.3507$) where layout geometry governs answerability. |
| Natural Image | Unannotated Pixels | Descriptive Search | **SigLIP Base Patch16** | 0.6071 [0.584, 0.633] | None (SigLIP is decisive; dominates CLIP and proxies) | Bypass text proxies completely; native visual embeddings dominate baseline BLIP (0.3728) and CLIP (0.4709); exploratory caption scaling is projected at $\leq 0.4420$. |
| Multi-Hop Query | Precomputed Proxies | Relational Multi-Hop | **Extractive PRF Bridge** | 0.8200 [0.742, 0.898]† | Extractive PRF (0.820), Generative LLM (0.820) | Extractive entity PRF matches generative LLM bridging on $N = 100$ relational queries ($p_{\text{TOST}} = 0.0073$ at $\Delta = \pm 0.05$), eliminating 394× latency multiplier. |

*Metric is MRR for rows 1–7; †Target Hit@10 on $N = 100$ relational queries ($N = 50$ pure 2-hop: 0.6400). Note: Comparisons with no statistically significant difference detected ($p > 0.05$) reflect failure to reject the null hypothesis; they do not assert statistical equivalence.

phrases (+0.0628 lift over Word BM25) or when visual layout governs answerability (DocVQA MRR 0.5497 vs. proxies $\leq$ 0.3507, noting in-domain training overlap). For answerable factual questions with preserved topical anchors, CPU lexical retrieval on real OCR matches or outperforms ColQwen2 (Char BM25 0.9328 vs. 0.8595, paired diff +0.0733, $p = 0.0011$; Clean GT 0.9471, diff +0.0876, $p = 0.00012$) while operating at 0.87 ms CPU lookup latency versus 249.21 ms GPU serving latency.

3) **For Photographic Archives:** Deploy native vision-language embeddings (SigLIP Base Patch16). Machine caption proxies suffer significant information loss (measured BLIP MRR 0.3728 vs. 0.6071, $p < 10^{-4}$; exploratory template scaling with Florence-2 is projected to reach 0.4420, retaining a −0.1651 deficit).
4) **For Multi-Hop Relational Reasoning:** Deploy rule-based extractive PRF over generative LLM bridging, matching retrieval reachability (0.640 vs. 0.640 Hit@10 on 50 hard queries, 0.820 vs. 0.820 on pooled 100 queries) with zero LLM cost.

## 6. NEGATIVE AND EXPLORATORY FINDINGS

To document architectural boundaries transparently, this section consolidates candidate mechanisms and feasibility pilots that failed to improve held-out performance.

### *A. Ablation of Regime-Aware Multimodal Proxy (RAMP) Hypotheses H1 and H2*

In exploratory testing, candidate mechanisms for adaptive routing and reliability weighting failed to outperform standard baselines:

1) **Hypothesis H1 (Reliability Weighting Rejected):** Weighting sparse streams by candidate OCR reliability ($r_{\text{ocr}}(d) = \bar{c}_{\text{ocr}} \cdot \rho_{\text{vocab}}$) degraded retrieval: 20% noise MRR dropped from 0.3694 to **0.3574** (diff CI $[-0.0222, -0.0025]$), and real degraded OCR dropped from 0.6684 to **0.6368** (raw $p = 0.0189$, $p_{\text{Holm}} = 0.0945$, non-significant after Holm adjustment, Table IV). Out-of-vocabulary ratios do not reliably differentiate optical blur from domain terminology.
2) **Hypothesis H2 (Learned Query Router Not Supported):** A lightweight classifier trained on surface cues $\phi(q)$ yielded only +0.0071 lift on synthetic noise (0.3765 vs. 0.3694, $p_{\text{Holm}} = 0.042$) and zero detectable lift under real Apple Vision OCR (0.8121 vs. 0.8121, $p_{\text{Holm}} = 1.0$) or degraded blur

(0.6667 vs. 0.6684, $p_{\text{Holm}} = 1.0$). Surface features cannot predict representation efficacy under real-world proxy distortions.

### B. Exploratory Multi-Hop Query Decomposition and End-to-End Task Auditing

On the pooled $N = 100$ relational multi-hop benchmark, extractive PRF and generative LLM achieve Hit@10 = 0.8200 (82/100; paired difference 0.0000 [95% CI: $-0.0392, +0.0392$], formal TOST equivalence at $\Delta = \pm 0.05$, $p_{\text{TOST}} = 0.0073 < 0.01$). The initial 50-query subset remains underpowered ($p_{\text{TOST}} = 0.1109$), so no equivalence claim is made for it; extractive PRF eliminates the 4,736.8 ms query decomposition latency of generative LLM bridging (394× slower than 12.0 ms extractive PRF).

In end-to-end document question answering ($N = 150$ queries over 100 SROIE receipts and 50 DocVQA pages) with frozen `gpt-4o-mini` [48], retrieval rank bounds downstream answer quality. String evaluation against ground-truth extraction fields confirms that ColQwen2's superior pooled retrieval MRR (0.6978 vs. 0.5934 for Char BM25) translates directly to downstream accuracy: Oracle context achieves ANLS = 0.5600 (Exact Match 0.5600), native ColQwen2 achieves ANLS = 0.3990 (EM 0.2933), and Char 3–5g BM25 achieves ANLS = 0.3246 (EM 0.2400). Retrieval omissions drive Exact Match to zero as generators refuse ("UNANSWERABLE") or hallucinate when targets are omitted.

To validate answer quality in user-facing multimodal retrieval, an independent dual-reviewer audit evaluated $N = 110$ generation responses from photo-album ($N = 60$) and relational ($N = 50$) benchmarks across a three-dimensional rubric. Blinded evaluators achieved 100% agreement on photo albums, while relational queries showed minor variance ($\kappa = 0.8054$ faithfulness, 0.8431 relevance, 0.9196 correctness). Across the pooled benchmark, inter-rater reliability was high: observed agreement was 95.5% for faithfulness ($\kappa = 0.8716$ [95% CI: 0.753, 0.974]), 96.4% for query relevance ($\kappa = 0.8926$ [95% CI: 0.772, 0.976]), and 98.2% for answer correctness ($\kappa = 0.9451$ [95% CI: 0.853, 1.000]; composite mean $\kappa = 0.9031$ [47]).

## 7. LIMITATIONS AND THREATS TO VALIDITY

To facilitate objective assessment, we document four key methodological boundaries and threats to validity:

1) **Domain Diversity and Model Pretraining Overlap:** Primary evaluations focus on retail receipts (SROIE, $N = 626$), administrative forms (FUNSD, $N = 100$), DocVQA ($N = 500$), and scenes (MS-COCO, $N = 1{,}000$). While spanning structured and scene domains, dense multi-column prose and non-Latin typographies warrant dedicated corpora. Crucially, `ColQwen2-v1.0` [12] incorporates DocVQA splits in its ViDoRe training mixture, while text proxies operate zero-shot; hence its DocVQA advantage (live measured 0.5497 vs. $\leq 0.3507$) represents an upper bound on in-domain transfer rather than zero-shot generalization. In addition, synthetic distractors in 500K scaling repeat vendor names, stressing inverted index capacity rather than open web retrieval.
2) **Optical Noise Simulation vs. Real-World Artifacts:** Our physical degradation suite combines synthetic character substitutions (CER 0.0–0.64) with optical Gaussian blur ($\sigma = 1.5$), bilinear downsampling, and JPEG compression. Authentic capture exhibits correlated artifacts—non-uniform lighting, perspective skew, paper creases, and thermal fade—inducing localized failures beyond uniform blur.
3) **Finite-Sample Marginal vs. PAC-Style Uncertainty Bounds:** Under marginal risk control $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$, split conformal calibration controls expected risk ($\mathbb{E}[L] \leq \alpha = 0.05$, empirical mean 0.0491 at a 78.4% escalation rate, $N = 2{,}754$, 200 splits). For high-probability PAC-style bounds, Risk-Controlling Prediction Sets (RCPS, $\delta = 0.10$) achieve mean test miss risk $0.0429 \leq 0.05$. Empirical coverage reaches 82.0% across 200 splits ($\hat{R}_{\text{test}} \leq 0.05$); the gap from nominal 90% reflects compound dispersion ($\sigma_{\text{total}} = 0.0076$), predicting $\approx$ 17.5% exceedance under normal theory.
4) **Hardware Serving Non-Equivalence and Latency Economics:** Latency profiles reflect heterogeneous deployment tiers (Apple M2 CPU for lexical indexes; Nvidia Tesla T4 GPU for multi-vector late interaction). Cascade latency (16.20 ms) is an analytical component-sum estimate (0.87 ms CPU pre-filtering + scaled GPU reranking) rather than a monolithic runtime. Late interaction also demands substantial multi-vector storage ($\sim$ 59 KB/page) scaling linearly with corpus size.

## 8. CONCLUSION AND FUTURE WORK

In this work, we established that intermediate textual proxies in multimodal document retrieval exhibit well-defined efficacy boundaries across 1,626 multimodal items: First, character 3–5gram inverted indexing provides a resilient sparse baseline, outperforming word tokenization under heavy optical blur (Apple Clean: 0.9436 vs. 0.8371; Degraded: 0.8248 vs. 0.6972; Tesseract: 0.8412 vs. 0.7364), reversing under entity substitutions (0.2427 vs. 0.4334). Second, dense bi-encoder collapse on document text stems from query-span formulation mismatch rather than sequence truncation; natural answerable questions elevate BGE-Base MRR from 0.3256 to 0.5263. Third, native visual models outperform machine captions on photos (SigLIP MRR 0.6071 vs. 0.3728, $p < 10^{-4}$); on visual pages (DocVQA), native ColQwen2 surpasses proxies (live measured 0.5497 vs. $\leq 0.3507$) as an in-domain upper bound; yet on answerable receipt queries, CPU lexical search (Char BM25 0.9328) beats ColQwen2 (0.8595, diff +0.0733, $p = 0.0011$) at 44× lower latency (5.72 ms vs. 249.21 ms).

As an empirical boundary, marginal conformal gating underperforms on semantic subgroups: entity rewrites experience miss risk $0.0627 > 0.05$ despite 58.5% aggregate coverage, while an equal-escalation fixed-margin baseline performs comparably (MRR 0.3579 vs. 0.3577). On 100 relational queries, extractive PRF statistically matches generative LLM bridging (Hit@10 0.820 vs. 0.820, $p_{\text{TOST}} = 0.0073 < 0.01$ at $\Delta = \pm 0.05$) while eliminating a 394× latency multiplier. Formalizing these trade-offs provides an engineering roadmap for efficient, trustworthy multimodal retrieval.

## Appendix A
## COMPREHENSIVE HYPOTHESIS TESTING FAMILIES

To ensure complete statistical traceability, Tables XI–XIII and the empirical results text evaluate 35 pre-specified hypotheses across six families under sequential Holm-Bonferroni correction ($\alpha = 0.05$, $N_{\text{perm}} = 100{,}000$): **(1) Family 1: Verbatim Sparse Optical Noise ($m = 6$):** Char vs. Word BM25 across Synthetic 20% ($p < 10^{-4}$), Apple Clean ($p < 10^{-4}$), Degraded Blur ($p < 10^{-4}$), Tesseract 5.5 ($p < 10^{-4}$), EasyOCR Clean ($p < 10^{-4}$), and Clean GT ($p < 10^{-4}$). All statistically significant; Char BM25 dominates. **(2) Family 2: ColQwen2 vs. Text Proxies ($m = 5$):** ColQwen2 vs. Word/Hybrid on verbatim (statistically indistinguishable, MRR 0.8417 vs. 0.8371, diff +0.0046, raw $p = 0.7420$, $p_{\text{Holm}} = 1.0000$); Char BM25 vs. ColQwen2 on Real-OCR answerable (raw $p = 0.0011$, $p_{\text{Holm}} = 0.0033$, Char BM25 wins +0.0733); ColQwen2 vs. Word/Hybrid on paraphrase (raw $p = 0.0072, 0.0095$, $p_{\text{Holm}} < 0.015$, ColQwen2 wins +0.0628); ColQwen2 vs. zero proxy baseline on DocVQA ($p < 10^{-4}$). **(3) Family 3: Native Vision vs. Captions on MS-COCO ($m = 8$):** SigLIP vs. BLIP MiniLM/BGE ($p < 10^{-4}$); CLIP vs. BLIP MiniLM/BGE ($p < 10^{-4}$); SigLIP vs. CLIP ($p < 10^{-4}$); SigLIP vs. Single Gold BGE ($p < 10^{-4}$); SigLIP vs. 4-Ref Gold BGE (diff +0.0236, raw $p = 0.0905$, $p_{\text{Holm}} = 0.1809$, tied); SigLIP vs. 4-Ref Gold Hybrid (diff −0.0007, raw $p = 0.9563$, $p_{\text{Holm}} = 0.9563$, tied). **(4) Family 4: RAMP Ablation Hypotheses ($m = 7$):** Char BM25 + Neural Cross-Encoder ($p_{\text{raw}} = 0.0001, p_{\text{Holm}} = 0.0007$, degrades −0.0262); BGE Reranker lift over Word BM25 ($p_{\text{raw}} = 0.0011, p_{\text{Holm}} = 0.0066$, significant); MiniLM Reranker lift ($p_{\text{raw}} = 0.0090, p_{\text{Holm}} = 0.0450 \leq 0.05$, statistically significant); H1 Reliability Weighting on blur ($p_{\text{raw}} = 0.0189, p_{\text{Holm}} = 0.0756$, non-significant); H2 Learned Router on 20% synthetic noise ($p_{\text{raw}} = 0.0420, p_{\text{Holm}} = 0.1260$, non-significant); H2 Learned Router on degraded blur ($p_{\text{Holm}} = 1.0000$); H2 Learned Router on Apple clean ($p_{\text{Holm}} = 1.0000$). **(5) Family 5: FUNSD Form Transfer ($m = 4$):** Char vs. Word BM25 on Apple paraphrase ($p_{\text{Holm}} = 0.0480$); Dense BGE vs. Word on EasyOCR paraphrase (raw $p = 0.018$, $p_{\text{Holm}} = 0.0540$, non-significant after correction); Dense BGE vs. Word on Apple paraphrase ($p_{\text{Holm}} = 0.168$); Char vs. Word on clean verbatim ($p = 0.3800$, tied). **(6) Family 6: Multi-Hop Relational Reasoning ($m = 5$):** Generative LLM vs. Sparse BM25 ($N = 50$, raw $p = 0.0215, p_{\text{Holm}} = 0.1075$, non-significant after family-wise multiplicity correction); LLM vs. Hybrid/Dense ($N = 50$, $p = 1.0000$, tied); Extractive PRF vs. LLM Bridge ($N = 50$ hard: $p_{\text{TOST}} = 0.1109$ at $\Delta = \pm 0.05$, underpowered; $N = 100$ pooled: $p_{\text{TOST}} = 0.0073$, equivalent). All per-query rankings and bootstrap distributions are archived in the supplementary materials.

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